Complex Numbers: The nth Roots of a Complex Number

Questions 1–25

Question 1. Find all the solutions to each of the following equations, giving your answer in the form $a + bi$ where a and b are constants:

a) $z^3 - 8i = 0$
b) $z^3 + 27 = 0$
c) $z^2 = 1 + \sqrt{3}i$
d) $z^4 + 16 = 0$

Question 2. Find all the solutions to each of the following equations, giving your answers in the exact form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $z^5 - 32i = 0$
b) $z^3 = 1 + \sqrt{3}i$
c) $z^4 = -2\sqrt{2} + 2\sqrt{2}i$
d) $z^8 = 8 + 8\sqrt{3}i$

Question 3. Find all the solutions to each of the following equations, giving your answer in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$:

a) $z^7 - 128i = 0$

b) $z^6 = -32 - 32\sqrt{3}i$

c) $2z^4 - 9 = 9\sqrt{3}i$

Question 4. Find the cube roots of $-2 + 2i$ giving your answers in the form $a + bi$ where a and b are real constants rounded to three significant figures.

Question 5. Evaluate each of the following, giving your answers in the form $a+bi$ where a and b are real constants rounded to three significant figures:

a) $\sqrt{-2+2i}$
b) $\sqrt{-2+4i}$
c) $\sqrt[4]{4-4i}$
d) $\sqrt[3]{4+4i}$

Question 6. Find the two roots of the equation $(z-i)^2 = -4+12i$ giving your answers in the form $a+bi$ where a and b are real constants rounded to three significant figures.

Question 7. Find all the solutions of $(z-2)^3 = 8i$ giving your answers in the form $a+bi$ where a and b are exact real constants.

Question 8. It is given for real constants a and b that a root of $z^3 = a+bi$ is $-3\sqrt{3}+3i$.

a) Find the values of a and b.

b) Find all the roots of $z^3 = a+bi$ giving your answer in the form $p+qi$ where p and q are real constants.

c) Explain why the complex roots do not come in conjugate pairs in this case.

Question 9. It is given for real constants a and b that a root of $z^4 = a + bi$ is $1 + i$.

a) Find the values of a and b.

b) Find all the roots of $z^4 = a + bi$ giving your answer in the form $p + qi$ where p and q are real constants.

c) Explain why the complex roots do come in conjugate pairs in this case.

Question 10. It is given that $z^3 = 5 + 5\sqrt{3}i$.

a) Find the solutions to the equation giving your answers in the form $10^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

b) Show that these roots satisfy the equation $w^9 + 1000 = 0$.

Question 11. Solve $z^8 = (2+2i)^4$ giving your answers in the form $2^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 12.

a) Write $1+i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Write $\sqrt{3}-i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

c) Determine the four roots of the equation $z^4 = (1+i)^2(\sqrt{3}-i)^2$ giving your answers in the form $2^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 13. It is given that $z = 5\sqrt{3} + 5i$.

a) Find z in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

b) Determine the values of w such that $w^4 = z^5$ giving your answers in the form $10^{5/4}(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

Question 14. Solve $z^{5/3} = -1 - \sqrt{3}i$ giving your answers in the form $2^k \mathrm{e}^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 15. Solve $z^{4/3} = \sqrt{2} + \sqrt{2}i$ giving your answers in the form $2^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 16. It is given that $z_1 = 1 + \sqrt{3}i$ and $z_2 = \sqrt{3} - i$.

a) Find z_1 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

b) Find z_2 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

c) Find all the values of w such that $w^6 = z_1^3 z_2^2$ giving your answers in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

Question 17.

a) Find the expansion of $(a-1)^3$.

b) Hence find the solutions to $z^3 - 3z^2 + 3z - 9 = 0$ giving your answers in the form $a + bi$ where a and b are real constants.

Question 18. Find all the six roots of $z^6 - 6z^5 + 15z^4 - 20z^3 + 15z^2 - 6z + 2 = 0$ giving your answers in the form $a + bi$ where a and b are real constants.

Question 19. It is given that $z = 1 + i$ and n is the smallest positive whole number such that z^n is a positive whole number.

a) Determine the value of n.
b) Find the value of z^n.
c) Write down the value of all roots of z^n for your value of n in the form $a + bi$ where a and b are real constants.

Question 20. It is given that $z = -7 + 7\sqrt{3}i$ and n is the smallest positive whole number such that z^n is a positive whole number.

a) Determine the value of n.
b) Find the value of z^n.
c) Write down all the roots of z^n for your value of n in the form $a + bi$ where a and b are real constants.

Question 21. It is given that $z = \frac{2\sqrt{3}+2i}{\sqrt{3}-i}$ and n is the smallest positive whole number such that z^n is a positive whole number.

a) Determine the value of n.

b) Find the value of z^n.

c) Write down all the roots of z^n for your value of n in the form $a + bi$ where a and b are real constants.

Question 22. Find all the solutions of $16z^4 = (z - i)^4$ giving your answers in the form $a + bi$ where a and b are exact real constants.

Question 23. It is given that $z^3 - 1 = \sqrt{3}i$.

a) Determine the three roots of the equation giving your answers in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Hence, show that $\cos\left(\frac{\pi}{9}\right) + \cos\left(\frac{5\pi}{9}\right) + \cos\left(\frac{7\pi}{9}\right) = 0$.

Question 24. It is given that $z^3 - 6\sqrt{3} = 6i$.

a) Determine the three roots of the equation giving your answers in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Hence, show that $\sin\left(\frac{\pi}{18}\right) - \sin\left(\frac{11\pi}{18}\right) + \sin\left(\frac{13\pi}{18}\right) = 0$.

Question 25. It is given that $z_1 = 1 + \sqrt{3}i$ and $z_2 = 1 - \sqrt{3}i$.

a) Write z_1 in the form $re^{i\theta_1}$ where $-\pi < \theta_1 \leq \pi$ and $r > 0$.

b) Write z_2 in the form $re^{i\theta_2}$ where $-\pi < \theta_2 \leq \pi$ and $r > 0$.

c) Show that $z_1^6 + z_2^6 = 128$ and $z_1^6 - z_2^6 = 0$. Given ω is a cube root of unity satisfying $\omega^3 = 1$ and $\omega \neq 1$, **d)** Evaluate and simplify $z_1^6 + \omega z_2^6 + \omega^2 (z_1 z_2)^6$ giving your answer in the form $a + bi$ where a and b are real constants.